Exact value triangles

1 Find the exact value of the following:

a $\sin 30^\circ$

b $\sin 60^\circ$

c $\sin 45^\circ$

d $\cos 45^\circ$

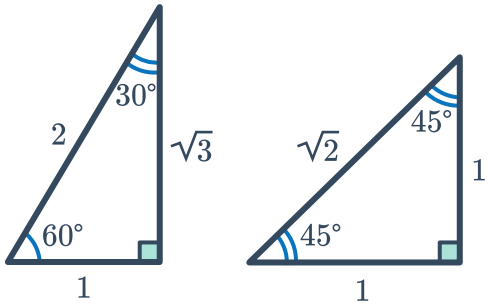
e $\cos 60^\circ$

f $\cos 30^\circ$

g $\tan 60^\circ$

h $\tan 45^\circ$

i $\tan 30^\circ$



2 Evaluate the following expressions, leaving your answer in exact, rationalised form:

a $\dfrac{\sin 60^\circ}{\cos 30^\circ}$

b $\sin 45^\circ \cos 30^\circ$

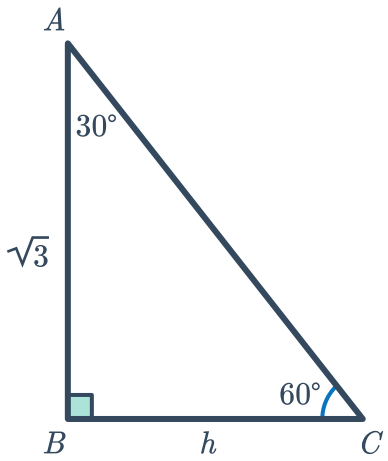
c $\sin 45^\circ + \cos 30^\circ$

d $\sin 45^\circ \cos 60^\circ + \tan 45^\circ$

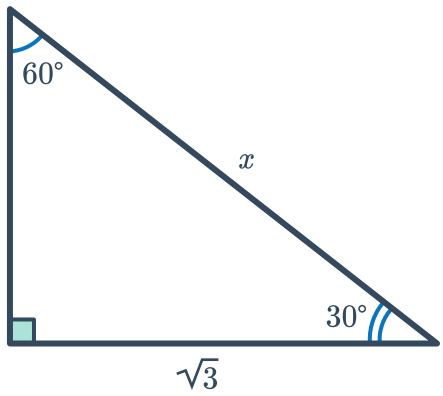
Unknown sides

3 Find the exact value of the pronumeral in the following triangles:

a

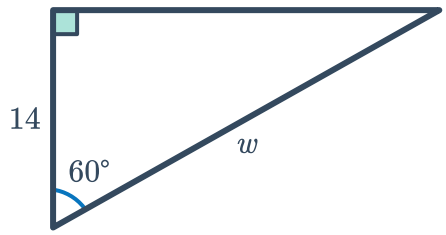


b

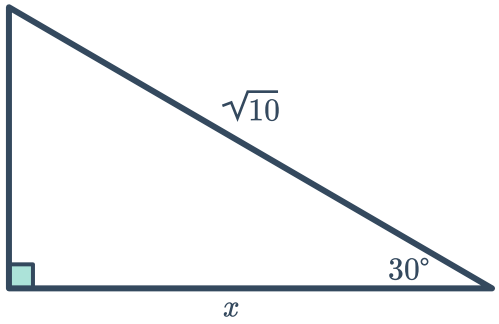


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c



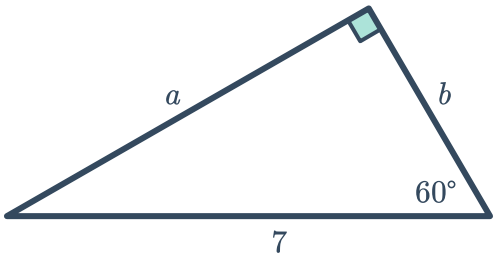
d



4 Consider the given triangle:

a Find the exact length of side a.

b Find the exact length of side b.

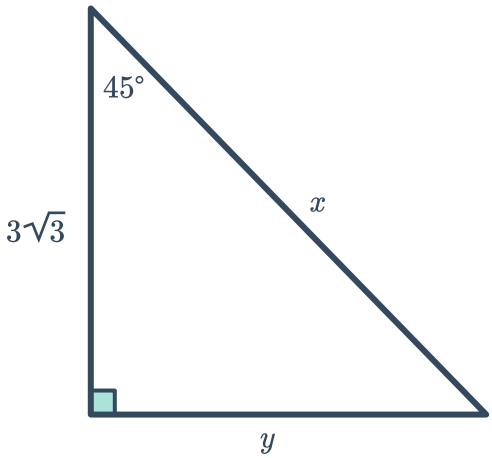


5 For each of the following triangles:

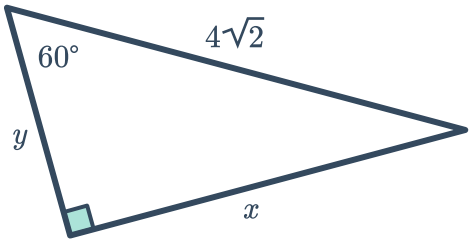
i Find the exact value of x.

ii Find the exact value of y.

a



b

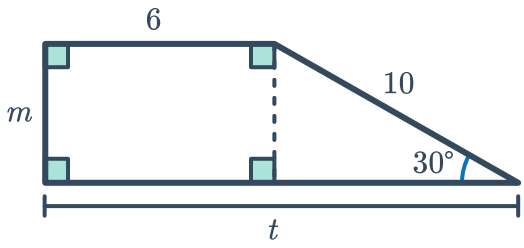


6 Consider the unknown lengths in the given figure:

a Find m.

b Find t.

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Unknown angles

7 θ is an angle in a right-angled triangle. Find the value of θ in the following equations:

a $\cos\theta = \frac{1}{2}$ **b** $\sin\theta = \frac{\sqrt{3}}{2}$ **c** $\sin\theta = \frac{1}{\sqrt{2}}$ **d** $\tan\theta = \frac{1}{\sqrt{3}}$

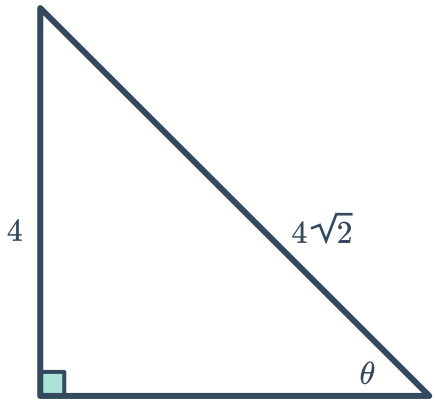
8 Find the value of θ if $\cos\theta = \frac{1}{\sqrt{2}}$ and $\sin\theta = \frac{1}{\sqrt{2}}$.

9 Given that $\cos\theta = \frac{\sqrt{3}}{2}$ and $\sin\theta = \frac{1}{2}$:

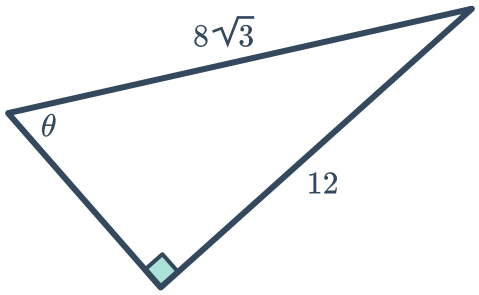
a Find the value of θ. **b** Find the value of $\tan\theta$.

10 Find the unknown θ in the following triangles:

a



b



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